

Sparse Regression of Power Amplifier Behavioral Models with a Stagewise Doubly Orthogonal Matching Pursuit

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Abstract—This work presents the sparse regression of Volterra-based behavioral models and digital predistorters (DPDs) through a Stagewise Doubly Orthogonal Matching Pursuit (StDOMP). First, the general sparse regression framework is presented. The StDOMP is formulated by applying the stagewise strategy to the DOMP algorithm, i.e., allowing the technique to select a block of components in each iteration without the need to perform one iteration per atom. The block selection approach allows to reduce the computational complexity up to a certain point where the performance is expected to be degraded. Experimental results highlight the behavior of the technique in the predistortion of a class AB power amplifier (PA) working in its nonlinear regime, concluding that the StDOMP is able to reduce runtime while still holding the performance level with respect to the DOMP.

I. INTRODUCTION

The design of power amplifiers (PAs) presents a challenge when it comes to the linearity versus efficiency trade-off. These devices show a nonlinear behavior at high powers, range where they are more efficient. Digital predistortion (DPD) raised as a flagship technique that allows to mitigate the nonlinear behavior of efficient PAs while working near their saturation zone. DPDs are often based on Volterra series structures that allow to mathematically model a nonlinear behavior with memory.

The high volume of coefficients in Volterra-based models renders compressed sensing techniques a convenient solution to reduce their computational complexity. For this matter, a plethora of techniques have been presented in the literature. Although their fundamental operation always follow the basic principle of selecting those coefficients that are more relevant to represent the output signal, every specific implementation shows subtle differences with regard to establishing the selection strategy.

The literature holds classical techniques, such as matching pursuit (MP), orthogonal MP (OMP), doubly OMP (DOMP) [1], sparse Bayesian learning (SBL) [2], dynamic model sizing (DMS) [3] and subspace pursuit (SP) that are based on matching pursuit schemes. One of the main problems that arise with the use of matching pursuit strategies such as the OMP and the DOMP is the need to perform at least as many iterations as there are atoms to be selected [4]. One of the possible solutions to this situation is the use of thresholding techniques like the Stagewise Orthogonal Matching Pursuit (StOMP) [5], that sets a threshold in the coefficient selection

operation of the algorithm to add several components per iteration that exhibit a correlation with the residual error larger than the threshold.

In this work we explore the application of the stagewise strategy to the DOMP algorithm. Instead of selecting one component per iteration, the Stagewise Doubly Orthogonal Matching Pursuit (StDOMP) will combine a block of components into the support set, consequently reducing the computational complexity of the overall technique. Since the operation that exhibits the highest computational complexity in the execution of the algorithm is the regression, it is expected to find a reduction in complexity proportional to the size of the block of components selected. On the other hand, after reducing the granularity at which the components are analyzed, it is also expected that the fidelity performance of the technique is degraded above some value of the block size. This effect is also investigated.

The rest of this work is organized as follows. First, Section II sets the general sparse regression framework. The StDOMP is presented in Section III and its computational complexity is reviewed in Section IV. Next, Section V covers the experimental design and the results are shown in Section VI. Section VII wraps up the work with the conclusions.

II. FRAMEWORK

The measurement equation relates the input complex envelope $\mathbf{x}$ to its output $\mathbf{y}$ through

$$\mathbf{y} = \mathbf{X}\mathbf{h} + \mathbf{e}, \quad (1)$$

where $\mathbf{y} \in \mathbb{C}^{m \times 1}$ is a stack of m samples of the input signal, $\mathbf{X} \in \mathbb{C}^{m \times n}$ is the model matrix that holds one regressor in each of its columns, $\mathbf{h} \in \mathbb{C}^{n \times 1}$ is the corresponding model coefficients vector and $\mathbf{e} \in \mathbb{C}^{m \times 1}$ accounts for the error.

The least squares (LS) method is able to attain an estimator of the coefficients vector $\hat{\mathbf{h}}$ with the regression

$$\hat{\mathbf{h}} = (\mathbf{X}^H \mathbf{X})^{-1} \mathbf{X}^H \mathbf{y} = \mathbf{X}^\dagger \mathbf{y}, \quad (2)$$

where $(\cdot)^H$, $(\cdot)^{-1}$, $(\cdot)^\dagger$ denote Hermitian, inverse, and pseudoinverse, respectively.

In the context of coefficient selection techniques, the support set S contains the indices of the regressors that are enabled or activated. Defining $\mathbf{X}_S$ as the operation of column selection,

$$\hat{\mathbf{h}}_S = \mathbf{X}_S^\dagger \mathbf{y}, \quad (3)$$

Algorithm 1 StDOMP pseudocode

Input: $\mathbf{X}, \mathbf{y}, B$
Output: $S^{(\text{end})}, \hat{\mathbf{h}}^{(\text{end})}$

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1: Initialization :  $\mathbf{r}^{(0)} \leftarrow \mathbf{y}, S^{(0)} \leftarrow \emptyset, \mathbf{Z}^{(0)} \leftarrow \mathbf{X}$ 
2: for iteration  $t = 1$  till stopping criterion is met do
3:    $\mathbf{g}_{\{i\}}^{(t)} \leftarrow \frac{\forall i \quad \mathbf{Z}_{\{i\}}^H}{\|\mathbf{Z}_{\{i\}}\|_2} \mathbf{r}^{(t-1)}$ 
4:    $s^{(t)} \leftarrow \text{supp} \left( H_B |\mathbf{g}^{(t)}| \right)$ 
5:    $S^{(t)} \leftarrow S^{(t-1)} \cup s^{(t)}$ 
6:   for  $i \in s^{(t)}$  do
7:      $\mathbf{p}^{(t)} \leftarrow \mathbf{Z}_{\{s^{(t)}\}}^{(t-1)H} \mathbf{Z}^{(t-1)}$ 
8:      $\mathbf{Z}^{(t)} \leftarrow \mathbf{Z}^{(t-1)} - \mathbf{p}^{(t)} \otimes \mathbf{Z}_{\{s^{(t)}\}}^{(t-1)}$ 
9:   end for
10:   $\hat{\mathbf{h}}_{S^{(t)}} \leftarrow \mathbf{X}_{S^{(t)}}^\dagger \mathbf{y}$ 
11:   $\hat{\mathbf{y}}^{(t)} \leftarrow \mathbf{X}_{S^{(t)}} \hat{\mathbf{h}}_{S^{(t)}}$ 
12:   $\mathbf{r}^{(t)} \leftarrow \mathbf{y} - \hat{\mathbf{y}}^{(t)}$ 
13: end for
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results in $\hat{\mathbf{h}}_S$ having S active or not null components. Correspondingly, $\hat{\mathbf{h}}_{\bar{S}} = \mathbf{0}$.

III. STAGewise DOMP (StDOMP)

This section is devoted to highlight the fundamental steps of the StDOMP algorithm. The DOMP algorithm tracks the information of the regressors in two spaces simultaneously: the Volterra space, denoted by $\mathbf{X}$ and an orthogonal space $\mathbf{Z}$. The orthogonal space $\mathbf{Z}$ is the result of performing Gram-Schmidt on the Volterra matrix $\mathbf{X}$ in the order of the support set. In its iterative execution, the algorithm includes B new regressors in the support set per iteration and monitors the part of the signal $\mathbf{y}$ that cannot be modeled with the residual error $\mathbf{r}$. Algorithm 1 shows the pseudocode of the StDOMP technique, which consists of:

A. Input parameters and initialization

The input variables to the block DOMP algorithm are the measurement matrix $\mathbf{X}$, the signal to model $\mathbf{y}$, and the block size B . Initialization is performed in step 1 of the algorithm. The residual equals the signal to model $\mathbf{y}$, and the support set S is empty in the first iteration since there are no regressors selected yet. Following the same logic, the orthogonal space in the first iteration is equal to the Volterra space.

B. Coefficient selection

The correlation $\mathbf{g}_{\{i\}}^{(t)}$ between the orthogonalized regressors and the residual error is calculated in step 3 of the algorithm. Please note that the regressors are normalized with the division by their ℓ_2 norm before the projection over the residual error to account for the scale differences.

Step 4 selects the B regressors that exhibit the highest correlation $|g|$ with the residual. This operation is performed through the nonlinear projection $H_k(\cdot)$, that sets all but the largest k elements of its argument to zero in conjunction with the support operation $\text{supp}(\cdot)$, that returns the indices of the elements of the argument that are not equal to zero. The

TABLE I
LINEARIZATION RESULTS OF THE StDOMP IN THE 7TH DLA ITERATION

Case	NMSE (dB)	ACPR _{-1/+1} (dBc)	EVM (%)	Normalized run time
w/o DPD	-24.6	-33.90 / -34.80	5.4	-
StDOMP $B = 1$	-38.2	-44.47 / -44.85	1.3	6.43
StDOMP $B = 5$	-37.7	-44.04 / -44.41	1.3	2.01
StDOMP $B = 10$	-37.7	-43.98 / -44.41	1.3	1.38
StDOMP $B = 25$	-37.4	-43.73 / -44.00	1.3	1.08
StDOMP $B = 50$	-37.1	-43.46 / -43.74	1.4	1

indices of basis functions that fulfill this condition are merged into the support set in step 5.

C. Orthogonalization process

The orthogonalization of the selected regressors with respect to the remaining of the Volterra space is carried out in steps 7 and 8 of the algorithm, where the operations are performed one by one over the set of B selected atoms at iteration t , indicated by the suffix $s_{\{i\}}^{(t)}$.

D. Estimations update

Finally, the algorithm updates the estimation of the Volterra kernel vector $\hat{\mathbf{h}}_{S^{(t)}}$, the estimated output $\hat{\mathbf{y}}^{(t)}$ and the residual error $\mathbf{r}^{(t)}$ in steps 10, 11, and 12, respectively.

IV. COMPUTATIONAL COMPLEXITY ASSESSMENT

In order to discuss the benefits of this new proposal, the computational complexity of the algorithm is evaluated in this section. The assessment is presented by using Bachmann–Landau notation in terms of the number of real-number multiplications $\mathcal{O}$. The principal advantage of StDOMP is expected to be evidenced by a reduction of the number of pseudoinverses performed. The computational complexity of the StDOMP technique follows

$$\mathcal{O}_{\text{StDOMP}} = \frac{n^4}{4B^4} + \left(\frac{4}{3B^3} - \frac{2}{B^2} + \frac{1}{B} \right) mn^3 + m^2n^2 + \left(\frac{5}{2B^2} - \frac{2}{B} + 1 \right) mn^2 + \frac{n^3}{2B^3}. \quad (4)$$

Please note that $\mathcal{O}_{\text{DOMP}} = \mathcal{O}_{\text{StDOMP}}|_{B=1}$. It can be appreciated how the computational complexity is the same in both methods, but with a reduction of $(1/B)$ on the highest orders in the case of StDOMP. The dimensions are $m > n$, but the exponent of n grows up quickly. The dominant term is m^2n^2 followed by $\left(\frac{4}{3B^3} - \frac{2}{B^2} + \frac{1}{B} \right) mn^3$ and $\frac{n^4}{4B^4}$ that helps to reduce the computational impact of the method. This complexity reduction is also proved by measuring runtime in Section VI.

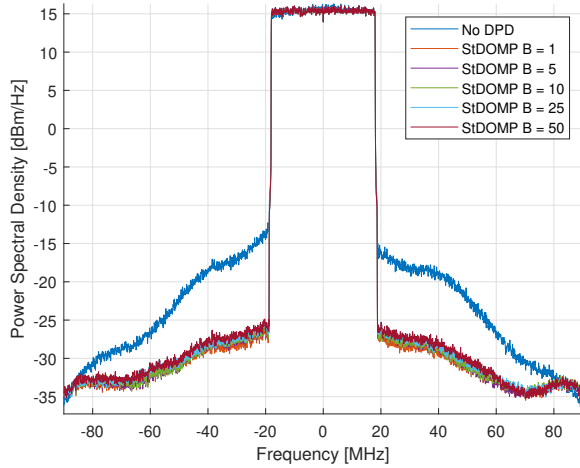


Fig. 1. Power spectral densities (PSDs) of the output signal without DPD and with DPD attained with the different block lengths by the StDOMP.

V. EXPERIMENTAL DESIGN

In order to highlight the above-mentioned features, the DPD of a commercial PA was carried out. The PA is based on the CGH40006-TB testboard for the GaN transistor CGH40006P from Cree, operated at center frequency 2.14 GHz. The experimental setup employed in this case was the RF WebLab maintained by Chalmers University of Technology [6], containing a Vector Signal Transceiver PXIe-5646R from National Instruments with 200 MHz instantaneous bandwidth, a PXI-4130 power source module from National Instruments providing the power supply, a linear driver amplifier with 40 dB gain, the PA under test, and a 30 dB attenuator.

The probing signal was an OFDM signal with 40 MHz bandwidth, 30 kHz separation between subcarriers, and 16-QAM symbols over all the subcarriers, following the 5G-NR format. It presented a PAPR of 10.5 dB. A sampling frequency of 184.32 MSa/s was employed both for the generation of the probing signal and for its acquisition at the output of the PA under test. For the operation point, the average output power was 26.6 dBm.

The selected model to perform the regression was a 412 coefficients model generated from a subset of Full Volterra regressors of order 18 and maximum memory of 50. From this model, only the first 200 regressors given by the StDOMP are used, easing the comparison between the different block lengths.

The DPD was performed following a direct learning architecture (DLA), enhanced by the usage of the StDOMP and normalization of its regressors. Seven iterations of the algorithm were conducted with a decaying learning rate to stabilize the system once the local minimum was found.

VI. RESULTS

In this section, the proposed technique is benchmarked regarding its linearization capabilities depending on the block size B . The block size value was swept in the values of 1

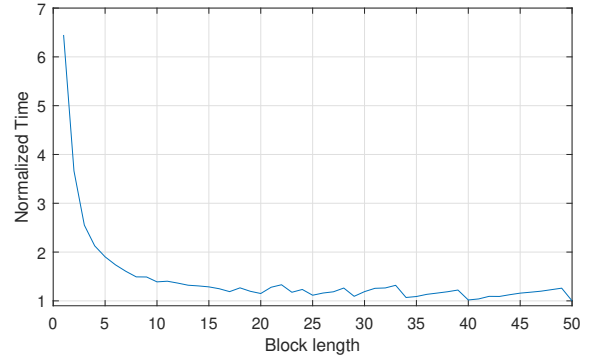


Fig. 2. Normalized runtime of the StDOMP technique against block size.

(this case is equal to the DOMP), 5, 10, 25 and 50 regressors per block. Table I shows the linearization achieved by these configurations. In this case, all of them perform similarly degrading its NMSE, ACPR and EVM inversely proportional to the length of the block. The most remarkable fact that can be observed is that the degradation shown by using the StDOMP over the classical DOMP is low.

To validate the results shown above, Fig. 1 presents the spectrum without DPD and with DPD attained with this technique in the above-mentioned configurations. The spectrum similarities validate the low performance loss after the increment of the block size.

Lastly, results regarding the execution time are presented. In Fig. 2, the normalized execution time is shown versus the length of the block in a sweep from 1 to 50. The hyperbolic-like shape resembles the behavior of (4). Although it is also seen in Table I, it is interesting to note that a decrease of up to 6 times the amount of time can be achieved with a slight loss in performance. There exists a saw-like behavior as the block size grows. This behavior is caused by the number of regressors, 200, being a multiple of the block length or not.

VII. CONCLUSION

In this work, the stagewise mechanism has been incorporated to the DOMP technique to perform an StDOMP algorithm. The selection of regressors in block is shown to reduce the computational complexity of the selection algorithm while keeping the linearization performance of its resulting model structures. Experimental results validate the computational complexity calculations and show the equivalent performance of solutions with different values of the block size. Performance degradation and its relation with the block size is expected to depend on the complexity of the nonlinear behavior of the PA, thus further work will include the validation of the technique in more PA classes.

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